

also be modified consistently because removing one output component changes the corresponding input dimension of the following operation. Figure 3.5 illustrates this difference between individual-weight removal and dimension-reducing structured pruning.[47, 48, 46]

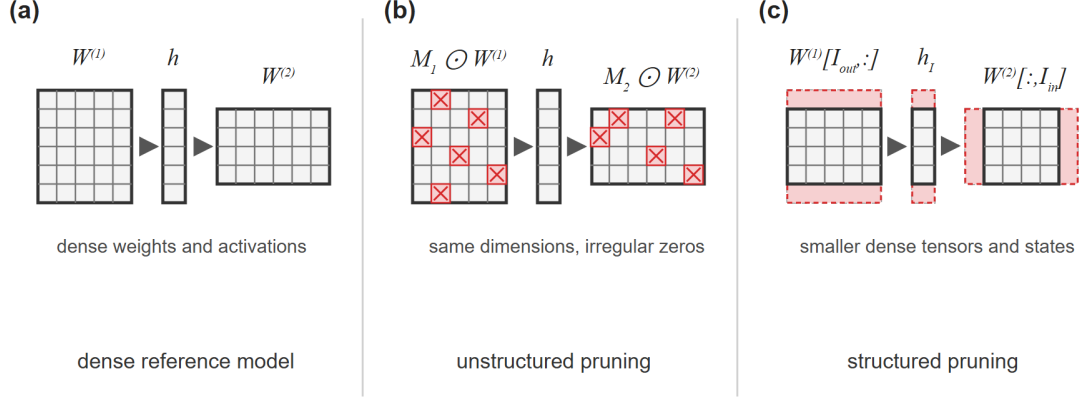


Figure 3.5: Comparison of pruning granularities. Panel (a) shows two dense weight matrices connected through an intermediate representation. In panel (b), unstructured pruning suppresses individual weights while retaining the original tensor dimensions. Panel (c) illustrates structured pruning, where complete output structures and the corresponding inputs of the following layer are removed. The resulting tensors and intermediate representation are smaller but remain dense.

Table 3.3 summarizes the practical trade-off. Unstructured pruning provides greater flexibility in selecting individual connections, while structured pruning more directly translates the removal decision into conventional embedded memory and compute reductions. Neither strategy guarantees a proportional reduction in measured latency because the realized gain still depends on storage format, kernel implementation, memory traffic, and the target processor.